

KIRTA: Knowledge, Inclusion & Real-Time Assistance*

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October 4, 2026

Annotation

The essence of the project

KIRTA is a voice-first daily-life AI companion for blind and low-vision people.

Target audience

More than 1 billion people experience sight loss; 43 million of them are blind [1]. KIRTA's primary target audience is blind and low-vision people. However, later expansion into people with other disabilities is possible.

Problem

Nowadays, people depend on smartphones for navigation, communication, shopping, studying, working, banking, entertainment, and other mundane tasks. People with blindness are no exception; for them, smartphones serve as a lifeline [2], replacing their sight. Yet using a phone requires them to hold it, orient themselves on the touchscreen, and ask other people for help when needed. Existing apps mostly work on request; for example, reading text from a photo/screen one at a time or helping with navigation. They also struggle with understanding users' intent [3].

Solution

Instead of holding the phone all the time, the phone is secured by a case with a strap. Its camera acts as the user's eyes, analyzing the surroundings with AI. Then the information is narrated to the user's headphones. Most of the time, the app is silent, breaking the silence only in critical moments, when the user is in danger, or on request. KIRTA is also expected to help with

*Semester project in Metódy inžinierskej práce, academic year 2026/27, supervisor: Ing. Vladimír Mlynarovič, PhD.

navigation, calls, messages, and alerts. More functions are to come later; for example, ordering goods and food, consent-based face recognition [4–6], tourist guidance, and integration into desktop operating systems. Security comes first, so everything involving biometric data and hazard detection will run locally, and payments are always confirmed by the user in the service’s own app. Other features use the cloud when the internet is available.

Monetization

Essential and local functions will be free. However, the price of the premium subscription will depend on the support of partners (banks, retailers, transit) and the state [7].

Literature

- [1] International Agency for the Prevention of Blindness. Global data. IAPB Vision Atlas, 2026.
- [2] S. S. Senjam and S. A. Primo. Challenges and enablers for smartphone use by persons with vision loss during the COVID-19 pandemic: A report of two case studies. *Frontiers in Public Health*, 10, 2022.
- [3] J. Xie, R. Yu, H. Zhang, S. M. Billah, S. Lee, and J. M. Carroll. Beyond visual perception: Insights from smartphone interaction of visually impaired users with large multimodal models. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, 2025.
- [4] European Parliament and Council of the European Union. Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). Official Journal of the European Union, L 119, 4.5.2016, pp. 1–88, 2016.
- [5] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L, 2024/1689, 12.7.2024, 2024.
- [6] T. Akter, T. Ahmed, A. Kapadia, and M. Swaminathan. Shared privacy concerns of the visually impaired and sighted bystanders with camera-based assistive technologies. *ACM Transactions on Accessible Computing*, 15:1–33, 2022.
- [7] European Parliament and Council of the European Union. Directive (EU) 2019/882 on the accessibility requirements for products and services (European Accessibility Act). Official Journal of the European Union, L 151, 7.6.2019, pp. 70–115, 2019.